Supplementary Figure 1B, SpatialPCA Results

The following analysis uses Visium V2 data from Sample P2 in the paper High-definition spatial transcriptomic profiling of immune cell populations in colorectal cancer.

```
library(rhdf5)

library(ggplot2)
library(fs)
library(glue)

library(SpatialPCA)
```

Load the processed Data

Load the processed Visium data used in Figure 1.

```
counts <- read_h5ad_matrix(pth_proc_adata)
```

```
## The count matrix has dimensions  18035 x 4269
```

```
var <- read_h5ad_meta(pth_proc_adata, "/var")
obs <- read_h5ad_meta(pth_proc_adata, "/obs")

nobs <- nrow(obs)
if (which(dim(counts) == nobs) == 1L) {
    counts <- t(counts)
}
```

Use same genes as in Figure 1

```
# genes3k <- read_csv("../data/visium_genes3k.csv")
genes3k <- readr::read_csv("../../visium_genes3k.csv", col_names=F, col_select=1)
genes3k <- genes3k$X1
```

Define spatialPCA object

```
spa <- new("SpatialPCA")
spa@counts <- counts[genes3k,]
spa@normalized_expr <- as.matrix(spa@counts)
spa@location <- as.matrix(obs[,c("X","Y")])
```

Run spatial PCA

First, compute kernel

```
spa <- SpatialPCA_buildKernel(
  spa,
  sparseKernel = TRUE
)
```

```
## ## Selected kernel type is:  gaussian
## Gene 8  : sample is too sparse to find TD
## Gene 19  : sample is too sparse to find TD
## Gene 24  : sample is too sparse to find TD
## Gene 85  : sample is too sparse to find TD
## Gene 98  : sample is too sparse to find TD
## Gene 111  : sample is too sparse to find TD
## Gene 137  : sample is too sparse to find TD
## Gene 142  : sample is too sparse to find TD
## Gene 195  : sample is too sparse to find TD
## Gene 246  : sample is too sparse to find TD
## Gene 247  : sample is too sparse to find TD
## Gene 248  : sample is too sparse to find TD
## Gene 249  : sample is too sparse to find TD
## Gene 251  : sample is too sparse to find TD
## Gene 253  : sample is too sparse to find TD
## Gene 268  : sample is too sparse to find TD
## Gene 280  : sample is too sparse to find TD
## Gene 289  : sample is too sparse to find TD
## Gene 305  : sample is too sparse to find TD
## Gene 309  : sample is too sparse to find TD
## Gene 322  : sample is too sparse to find TD
## Gene 334  : sample is too sparse to find TD
## Gene 398  : sample is too sparse to find TD
## Gene 403  : sample is too sparse to find TD
## Gene 411  : sample is too sparse to find TD
## Gene 427  : sample is too sparse to find TD
## Gene 435  : sample is too sparse to find TD
## Gene 447  : sample is too sparse to find TD
## Gene 448  : sample is too sparse to find TD
## Gene 460  : sample is too sparse to find TD
## Gene 467  : sample is too sparse to find TD
## Gene 469  : sample is too sparse to find TD
## Gene 517  : sample is too sparse to find TD
## Gene 524  : sample is too sparse to find TD
## Gene 527  : sample is too sparse to find TD
## Gene 567  : sample is too sparse to find TD
## Gene 571  : sample is too sparse to find TD
## Gene 607  : sample is too sparse to find TD
## Gene 668  : sample is too sparse to find TD
## Gene 699  : sample is too sparse to find TD
## Gene 714  : sample is too sparse to find TD
## Gene 721  : sample is too sparse to find TD
## Gene 730  : sample is too sparse to find TD
## Gene 742  : sample is too sparse to find TD
## Gene 743  : sample is too sparse to find TD
## Gene 745  : sample is too sparse to find TD
## Gene 747  : sample is too sparse to find TD
## Gene 765  : sample is too sparse to find TD
## Gene 767  : sample is too sparse to find TD
## Gene 774  : sample is too sparse to find TD
## Gene 779  : sample is too sparse to find TD
## Gene 781  : sample is too sparse to find TD
## Gene 791  : sample is too sparse to find TD
## Gene 801  : sample is too sparse to find TD
## Gene 816  : sample is too sparse to find TD
## Gene 828  : sample is too sparse to find TD
## Gene 841  : sample is too sparse to find TD
## Gene 843  : sample is too sparse to find TD
## Gene 860  : sample is too sparse to find TD
## Gene 863  : sample is too sparse to find TD
## Gene 887  : sample is too sparse to find TD
## Gene 892  : sample is too sparse to find TD
## Gene 920  : sample is too sparse to find TD
## Gene 922  : sample is too sparse to find TD
## Gene 924  : sample is too sparse to find TD
## Gene 926  : sample is too sparse to find TD
## Gene 941  : sample is too sparse to find TD
## Gene 974  : sample is too sparse to find TD
## Gene 975  : sample is too sparse to find TD
## Gene 1039  : sample is too sparse to find TD
## Gene 1072  : sample is too sparse to find TD
## Gene 1132  : sample is too sparse to find TD
## Gene 1133  : sample is too sparse to find TD
## Gene 1163  : sample is too sparse to find TD
## Gene 1173  : sample is too sparse to find TD
## Gene 1199  : sample is too sparse to find TD
## Gene 1224  : sample is too sparse to find TD
## Gene 1229  : sample is too sparse to find TD
## Gene 1247  : sample is too sparse to find TD
## Gene 1260  : sample is too sparse to find TD
## Gene 1267  : sample is too sparse to find TD
## Gene 1285  : sample is too sparse to find TD
## Gene 1293  : sample is too sparse to find TD
## Gene 1296  : sample is too sparse to find TD
## Gene 1307  : sample is too sparse to find TD
## Gene 1323  : sample is too sparse to find TD
## Gene 1339  : sample is too sparse to find TD
## Gene 1368  : sample is too sparse to find TD
## Gene 1375  : sample is too sparse to find TD
## Gene 1427  : sample is too sparse to find TD
## Gene 1428  : sample is too sparse to find TD
## Gene 1490  : sample is too sparse to find TD
## Gene 1507  : sample is too sparse to find TD
## Gene 1509  : sample is too sparse to find TD
## Gene 1530  : sample is too sparse to find TD
## Gene 1556  : sample is too sparse to find TD
## Gene 1583  : sample is too sparse to find TD
## Gene 1593  : sample is too sparse to find TD
## Gene 1606  : sample is too sparse to find TD
## Gene 1621  : sample is too sparse to find TD
## Gene 1640  : sample is too sparse to find TD
## Gene 1647  : sample is too sparse to find TD
## Gene 1648  : sample is too sparse to find TD
## Gene 1650  : sample is too sparse to find TD
## Gene 1679  : sample is too sparse to find TD
## Gene 1697  : sample is too sparse to find TD
## Gene 1703  : sample is too sparse to find TD
## Gene 1732  : sample is too sparse to find TD
## Gene 1757  : sample is too sparse to find TD
## Gene 1834  : sample is too sparse to find TD
## Gene 1838  : sample is too sparse to find TD
## Gene 1871  : sample is too sparse to find TD
## Gene 1876  : sample is too sparse to find TD
## Gene 1895  : sample is too sparse to find TD
## Gene 1960  : sample is too sparse to find TD
## Gene 1973  : sample is too sparse to find TD
## Gene 1997  : sample is too sparse to find TD
## Gene 2007  : sample is too sparse to find TD
## Gene 2020  : sample is too sparse to find TD
## Gene 2035  : sample is too sparse to find TD
## Gene 2056  : sample is too sparse to find TD
## Gene 2081  : sample is too sparse to find TD
## Gene 2088  : sample is too sparse to find TD
## Gene 2098  : sample is too sparse to find TD
## Gene 2107  : sample is too sparse to find TD
## Gene 2110  : sample is too sparse to find TD
## Gene 2113  : sample is too sparse to find TD
## Gene 2121  : sample is too sparse to find TD
## Gene 2154  : sample is too sparse to find TD
## Gene 2156  : sample is too sparse to find TD
## Gene 2162  : sample is too sparse to find TD
## Gene 2175  : sample is too sparse to find TD
## Gene 2193  : sample is too sparse to find TD
## Gene 2197  : sample is too sparse to find TD
## Gene 2244  : sample is too sparse to find TD
## Gene 2249  : sample is too sparse to find TD
## Gene 2254  : sample is too sparse to find TD
## Gene 2255  : sample is too sparse to find TD
## Gene 2300  : sample is too sparse to find TD
## Gene 2311  : sample is too sparse to find TD
## Gene 2345  : sample is too sparse to find TD
## Gene 2378  : sample is too sparse to find TD
## Gene 2390  : sample is too sparse to find TD
## Gene 2397  : sample is too sparse to find TD
## Gene 2420  : sample is too sparse to find TD
## Gene 2422  : sample is too sparse to find TD
## Gene 2433  : sample is too sparse to find TD
## Gene 2439  : sample is too sparse to find TD
## Gene 2451  : sample is too sparse to find TD
## Gene 2456  : sample is too sparse to find TD
## Gene 2470  : sample is too sparse to find TD
## Gene 2481  : sample is too sparse to find TD
## Gene 2489  : sample is too sparse to find TD
## Gene 2526  : sample is too sparse to find TD
## Gene 2561  : sample is too sparse to find TD
## Gene 2566  : sample is too sparse to find TD
## Gene 2621  : sample is too sparse to find TD
## Gene 2640  : sample is too sparse to find TD
## Gene 2649  : sample is too sparse to find TD
## Gene 2658  : sample is too sparse to find TD
## Gene 2660  : sample is too sparse to find TD
## Gene 2706  : sample is too sparse to find TD
## Gene 2719  : sample is too sparse to find TD
## Gene 2724  : sample is too sparse to find TD
## Gene 2738  : sample is too sparse to find TD
## Gene 2751  : sample is too sparse to find TD
## Gene 2754  : sample is too sparse to find TD
## Gene 2765  : sample is too sparse to find TD
## Gene 2769  : sample is too sparse to find TD
## Gene 2826  : sample is too sparse to find TD
## Gene 2897  : sample is too sparse to find TD
## Gene 2928  : sample is too sparse to find TD
## Gene 2929  : sample is too sparse to find TD
## Gene 2933  : sample is too sparse to find TD
## Gene 2937  : sample is too sparse to find TD
## Gene 2944  : sample is too sparse to find TD
## Gene 2948  : sample is too sparse to find TD
## Gene 2960  : sample is too sparse to find TD
## Gene 2962  : sample is too sparse to find TD
## Gene 2974  : sample is too sparse to find TD
## Gene 2977  : sample is too sparse to find TD
## Gene 2983  : sample is too sparse to find TD
## Gene 2989  : sample is too sparse to find TD
## Gene 2995  : sample is too sparse to find TD
## ## The bandwidth is:  0.0486129202344738
## ## Calculating sparse kernel matrix
## ## Finished calculating kernel matrix.
```

Now, run method

```
spa <- SpatialPCA_EstimateLoading(
  spa, fast=TRUE, SpatialPCnum=20)
```

```
## [1] "Low rank approximation!"
## [1] "Small sample, using top 195 eigenvectors and eigenvalues in the Kernel matrix!"
```

```
spa <- SpatialPCA_SpatialPCs(spa, fast=TRUE)
```

```
## [1] "Small sample, using top 195 eigenvectors and eigenvalues in the Kernel matrix!"
```

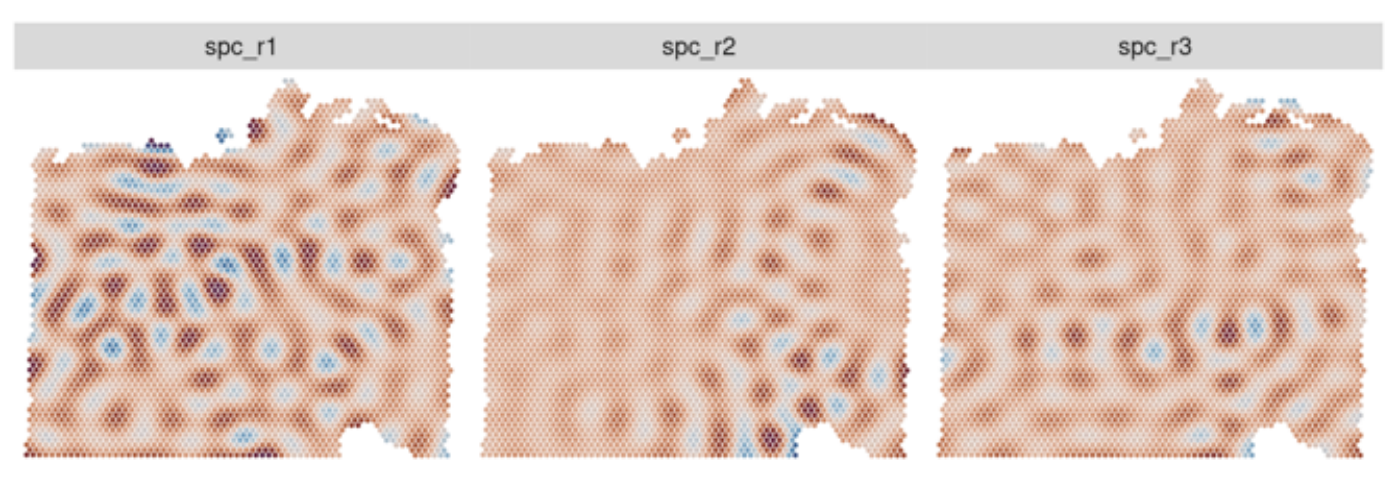
Aggregate results for plotting

```
spcs <- t(spa@SpatialPCs)
colnames(spcs) <- paste0("spc_r",seq(20))
obs <- cbind(obs, spcs)
```

Plot results

```
# Update for each spatial PC
pl_obs <- obs |>
  dplyr::select(X, Y, spc_r1:spc_r3) |>
  tidyr::pivot_longer(
    contains("spc"),
    names_to = "component",
    values_to = "score"
  )

ggplot(pl_obs, aes(X,-1*Y, color=score)) +
    geom_point(size=rel(0.25), alpha=0.90) +
    facet_wrap(~ component) +
    scale_x_continuous(expand=c(0.03,0.03)) +
    scale_y_continuous(expand=c(0.03,0.03)) +
    scico::scale_color_scico(
      breaks = function(x) range(x),
      labels = c("Low", "High"),
      palette = 'vikO', direction=1,
    ) +
  theme_krpca()
```



Session information

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: x86_64-pc-linux-gnu
## Running under: CentOS Stream 8
## 
## Matrix products: default
## BLAS:   /opt/R/4.5.1/lib64/R/lib/libRblas.so 
## LAPACK: /opt/R/4.5.1/lib64/R/lib/libRlapack.so;  LAPACK version 3.12.1
## 
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C              
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8    
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8   
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C                 
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C            
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C       
## 
## time zone: America/Los_Angeles
## tzcode source: system (glibc)
## 
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base     
## 
## other attached packages:
## [1] RSpectra_0.16-2  SpatialPCA_1.3.0 glue_1.8.0       fs_1.6.6        
## [5] ggplot2_4.0.1    rhdf5_2.54.1    
## 
## loaded via a namespace (and not attached):
##   [1] RColorBrewer_1.1-3     rstudioapi_0.17.1      jsonlite_2.0.0        
##   [4] umap_0.2.10.0          magrittr_2.0.4         spatstat.utils_3.2-1  
##   [7] farver_2.1.2           rmarkdown_2.30         vctrs_0.7.1           
##  [10] ROCR_1.0-11            spatstat.explore_3.6-0 askpass_1.2.1         
##  [13] CompQuadForm_1.4.4     htmltools_0.5.9        SPARK_1.1.1           
##  [16] Rhdf5lib_1.32.0        sass_0.4.10            sctransform_0.4.3     
##  [19] parallelly_1.46.1      pracma_2.4.6           KernSmooth_2.23-26    
##  [22] bslib_0.9.0            htmlwidgets_1.6.4      ica_1.0-3             
##  [25] plyr_1.8.9             plotly_4.11.0          zoo_1.8-15            
##  [28] cachem_1.1.0           igraph_2.2.0           mime_0.13             
##  [31] lifecycle_1.0.5        iterators_1.0.14       pkgconfig_2.0.3       
##  [34] Matrix_1.7-4           R6_2.6.1               fastmap_1.2.0         
##  [37] fitdistrplus_1.2-4     future_1.69.0          shiny_1.12.1          
##  [40] digest_0.6.39          patchwork_1.3.2        Seurat_5.4.0          
##  [43] tensor_1.5.1           irlba_2.3.5.1          pdist_1.2.1           
##  [46] labeling_0.4.3         progressr_0.18.0       spatstat.sparse_3.1-0 
##  [49] httr_1.4.7             polyclip_1.10-7        abind_1.4-8           
##  [52] compiler_4.5.1         bit64_4.6.0-1          withr_3.0.2           
##  [55] doParallel_1.0.17      S7_0.2.1               fastDummies_1.7.5     
##  [58] MASS_7.3-65            openssl_2.3.4          tools_4.5.1           
##  [61] lmtest_0.9-40          otel_0.2.0             httpuv_1.6.16         
##  [64] future.apply_1.20.1    goftest_1.2-3          nlme_3.1-168          
##  [67] rhdf5filters_1.22.0    promises_1.5.0         grid_4.5.1            
##  [70] Rtsne_0.17             cluster_2.1.8.1        reshape2_1.4.5        
##  [73] generics_0.1.4         gtable_0.3.6           spatstat.data_3.1-9   
##  [76] tzdb_0.5.0             tidyr_1.3.2            hms_1.1.4             
##  [79] data.table_1.18.0      sp_2.2-0               spatstat.geom_3.6-1   
##  [82] RcppAnnoy_0.0.23       ggrepel_0.9.6          RANN_2.6.2            
##  [85] foreach_1.5.2          pillar_1.11.1          stringr_1.6.0         
##  [88] vroom_1.7.0            spam_2.11-3            RcppHNSW_0.6.0        
##  [91] later_1.4.5            splines_4.5.1          dplyr_1.2.0           
##  [94] lattice_0.22-7         bit_4.6.0              survival_3.8-3        
##  [97] deldir_2.0-4           tidyselect_1.2.1       miniUI_0.1.2          
## [100] pbapply_1.7-4          knitr_1.51             gridExtra_2.3         
## [103] scattermore_1.2        xfun_0.56              matrixStats_1.5.0     
## [106] stringi_1.8.7          lazyeval_0.2.2         yaml_2.3.12           
## [109] evaluate_1.0.5         codetools_0.2-20       tibble_3.3.1          
## [112] cli_3.6.5              uwot_0.2.4             matlab_1.0.4.1        
## [115] xtable_1.8-4           reticulate_1.44.1      jquerylib_0.1.4       
## [118] dichromat_2.0-0.1      Rcpp_1.1.1             globals_0.18.0        
## [121] spatstat.random_3.4-3  png_0.1-8              spatstat.univar_3.1-6 
## [124] parallel_4.5.1         readr_2.1.6            dotCall64_1.2         
## [127] listenv_0.10.0         viridisLite_0.4.2      scales_1.4.0          
## [130] ggridges_0.5.7         crayon_1.5.3           SeuratObject_5.3.0    
## [133] purrr_1.2.1            scico_1.5.0            rlang_1.1.7           
## [136] formatR_1.14           cowplot_1.2.0
```